

Diabetes Risk Analysis

Exploratory Data Analysis & Statistical Hypothesis Testing on the Pima
Indians Diabetes Dataset

A Data Analytics Project Report

Dataset Source: National Institute of Diabetes and Digestive and Kidney Diseases (via UCI / Kaggle)

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy (Google Colab)

Date: July 2026

1. Project Overview

This project performs an exploratory data analysis (EDA) and statistical hypothesis testing on the Pima Indians Diabetes dataset, originally collected by the National Institute of Diabetes and Digestive and Kidney Diseases. The dataset contains diagnostic medical measurements for 768 female patients of Pima Indian heritage, all aged 21 years or older, with the goal of diagnostically predicting whether or not a patient has diabetes.

The dataset includes eight predictor (independent) variables and one target (dependent) variable, Outcome, which indicates whether a patient was diagnosed with diabetes (1) or not (0).

1.1 Project Title

"Diabetes Risk Analysis: Exploratory Data Analysis and Statistical Hypothesis Testing on the Pima Indians Diabetes Dataset"

1.2 Objective

The primary objective of this project is to explore, visualize, and statistically evaluate the relationships between key diagnostic health indicators and diabetes outcome, in order to identify which factors are most strongly and significantly associated with the onset of diabetes. Specifically, the project aims to:

- Understand the structure, quality, and distribution of the dataset.
- Visualize the distribution of key clinical variables such as Glucose, BMI, and Age.
- Detect outliers and assess data quality issues (e.g., biologically implausible zero values).
- Examine relationships and correlations between predictor variables and the diabetes Outcome.
- Apply statistical hypothesis tests (independent t-tests and one-way ANOVA) to determine which variables are significantly associated with diabetes.
- Summarize findings that could inform feature selection for a future predictive (machine learning) model.

1.3 Dataset Description

Column	Description
Pregnancies	Number of times pregnant
Glucose	Plasma glucose concentration (2-hour oral glucose tolerance test)
BloodPressure	Diastolic blood pressure (mm Hg)
SkinThickness	Triceps skin fold thickness (mm)
Insulin	2-Hour serum insulin (mu U/ml)
BMI	Body mass index (weight in kg / (height in m) ²)
DiabetesPedigreeFunction	A function scoring likelihood of diabetes based on family history
Age	Age in years
Outcome	Class variable: 1 = diabetic, 0 = non-diabetic (target)

The dataset has 768 rows and 9 columns, with no missing (null) values reported by pandas. However, as discussed in Section 3, several columns contain zero values that are not physiologically plausible (e.g., zero Glucose or zero BMI), which likely represent disguised missing data rather than true zeros.

2. Methodology

The analysis was carried out in Python using Google Colab, leveraging Pandas and NumPy for data handling, Matplotlib and Seaborn for visualization, and SciPy's stats module for hypothesis testing. The workflow followed these stages:

- Data loading and structural inspection (shape, columns, data types, missing values).
- Descriptive statistics (`df.describe()`) to understand central tendency and spread.
- Univariate visualization (histograms with KDE) of key numeric features.
- Outlier detection using boxplots (global and individual).
- Bivariate visualization (scatter plots) to examine relationships against Glucose and Outcome.
- Multivariate visualization (pair plot) segmented by Outcome.
- Correlation analysis using Pearson correlation coefficients and a heatmap.
- Class balance inspection (count plot and pie chart of Outcome).
- Distributional comparison (violin plot) of Glucose by diabetes status.
- Statistical hypothesis testing: independent-samples t-tests for Glucose, BMI, Blood Pressure, Age, and Pregnancies across Outcome groups; one-way ANOVA of Glucose across Age bands; and a Shapiro-Wilk normality test on Glucose.

3. Data Quality Check

An initial check with `df.isnull().sum()` confirmed there are no explicit missing values across any of the 9 columns. However, the descriptive statistics table (`df.describe()`) revealed minimum values of 0 for Glucose, BloodPressure, SkinThickness, Insulin, and BMI — values that are not medically possible for a living patient.

Interpretation: The presence of zero values in clinically essential fields (e.g., 0 mg/dL glucose or 0 BMI) strongly suggests these are placeholder values for missing/unrecorded measurements rather than genuine zeros. This is an important caveat: subsequent visualizations (especially the boxplots and scatter plots) show a cluster of points at zero for these variables, which slightly distorts distribution shapes and could bias statistical results if not addressed in a full modeling pipeline (e.g., via imputation).

4. Univariate Analysis — Distributions

4.1 Glucose Distribution

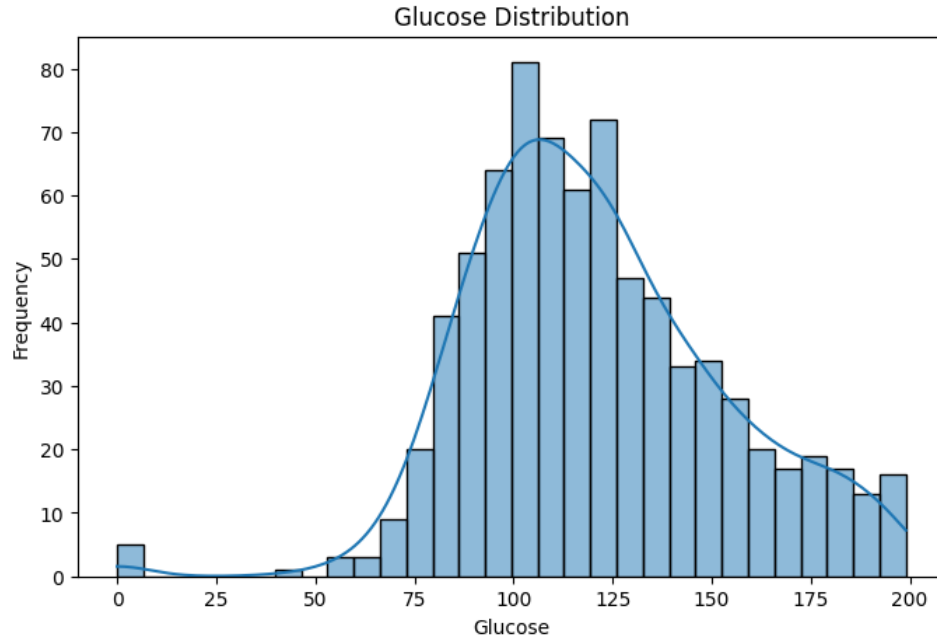


Figure 1: Distribution of Glucose levels (histogram with KDE overlay)

Interpretation: Glucose is approximately normally distributed with a slight right skew, centered around 110–120 mg/dL. A small spike near zero is visible, corresponding to the disguised missing values discussed in Section 3. Excluding that artifact, the bulk of readings fall within a physiologically typical range, with a long right tail extending toward 190–200 mg/dL representing patients with elevated glucose readings.

4.2 BMI Distribution

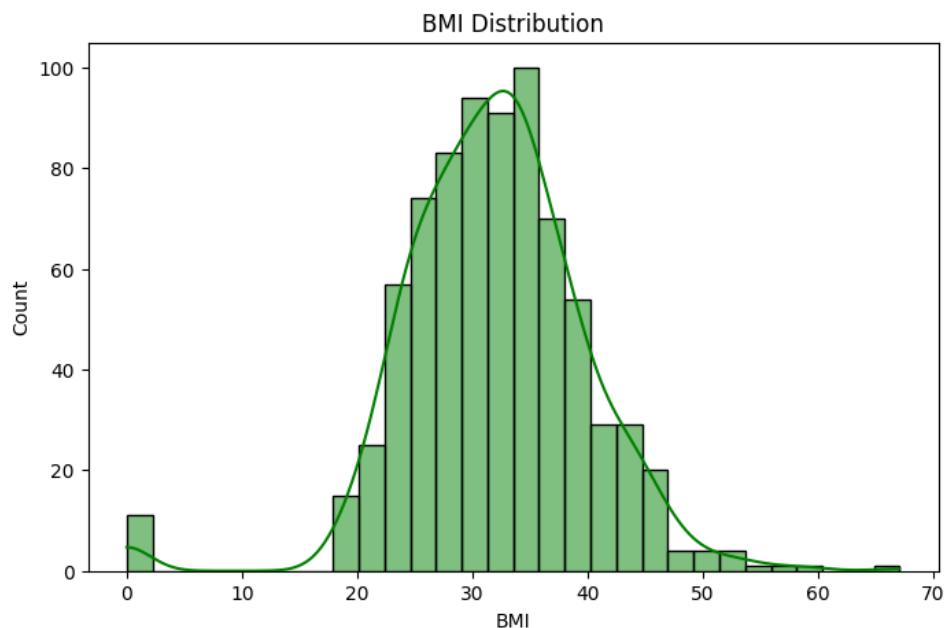


Figure 2: Distribution of Body Mass Index (BMI)

Interpretation: BMI follows a roughly bell-shaped (normal-like) distribution centered around 30–33, which falls in the "obese" range by WHO classification, indicating that the sample population skews toward higher body weight overall. As with Glucose, a small artificial spike at zero reflects missing/unrecorded BMI values rather than true zero BMI.

4.3 Age Distribution

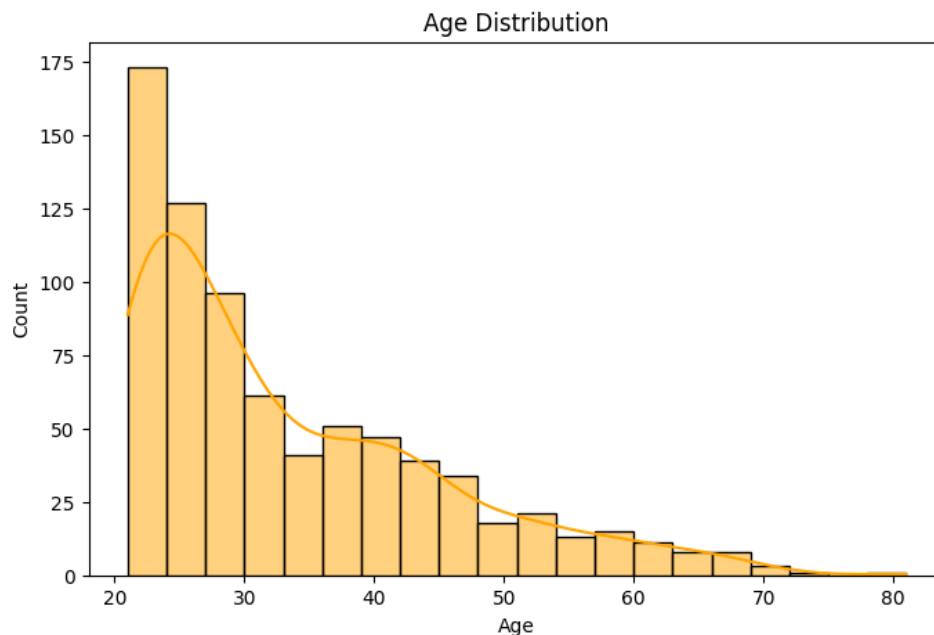


Figure 3: Distribution of patient Age

Interpretation: The age distribution is heavily right-skewed, with the largest concentration of patients between 21–30 years old and a steadily decreasing number of patients at older ages. This reflects the demographic composition of the sample rather than the general population, and means results for older age groups are based on comparatively fewer observations.

5. Outlier Detection

5.1 Combined Boxplot — All Features

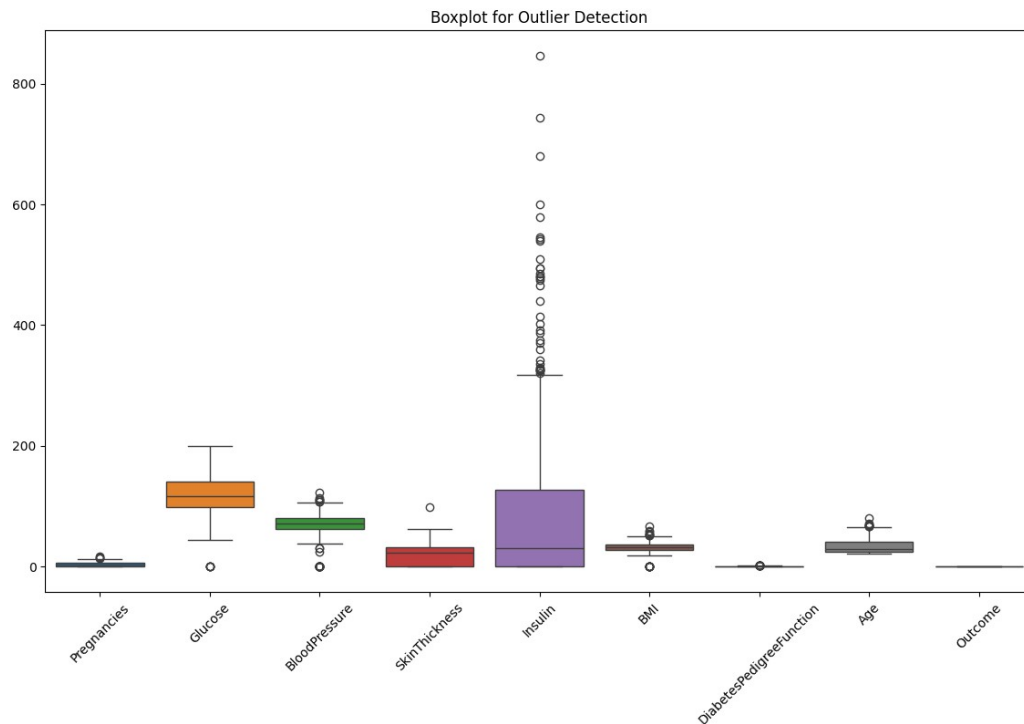


Figure 4: Boxplots of all numeric features (outlier detection)

Interpretation: Insulin shows the most extreme and numerous outliers, with values extending well beyond 500–800 mu U/ml, far above the interquartile range. SkinThickness and BMI also show a moderate number of high-end outliers. Pregnancies, BloodPressure, and DiabetesPedigreeFunction show comparatively few and modest outliers. The scale differences across variables (e.g., Insulin ranging into the hundreds versus DiabetesPedigreeFunction ranging near 0–2.5) illustrate why feature scaling/normalization would be necessary before using these variables in a distance-based machine learning model.

5.2 Glucose — Individual Boxplot

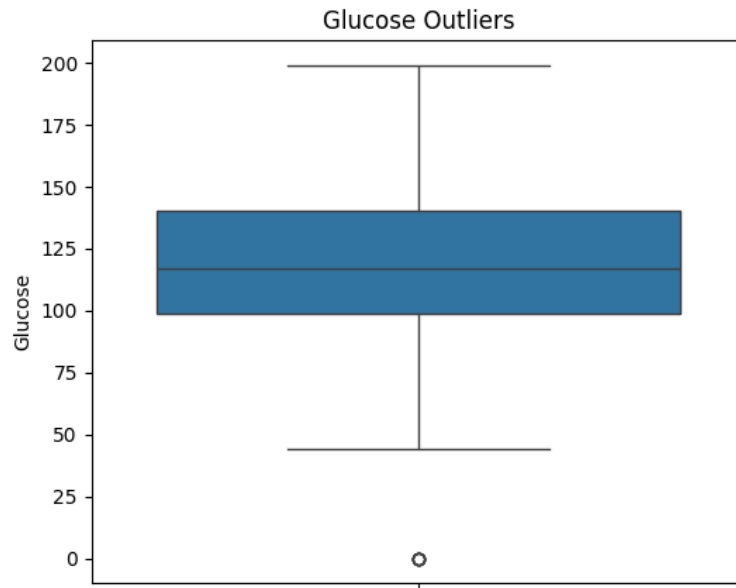


Figure 5: Boxplot of Glucose values highlighting the zero-value outlier

Interpretation: Isolating Glucose confirms a tight, fairly symmetric interquartile range (roughly 99–140), with the only outlier being the cluster of biologically impossible zero readings sitting far below the lower whisker — reinforcing that these should be treated as missing data rather than genuine low glucose measurements.

6. Bivariate & Multivariate Analysis

6.1 BMI vs Glucose (colored by Outcome)

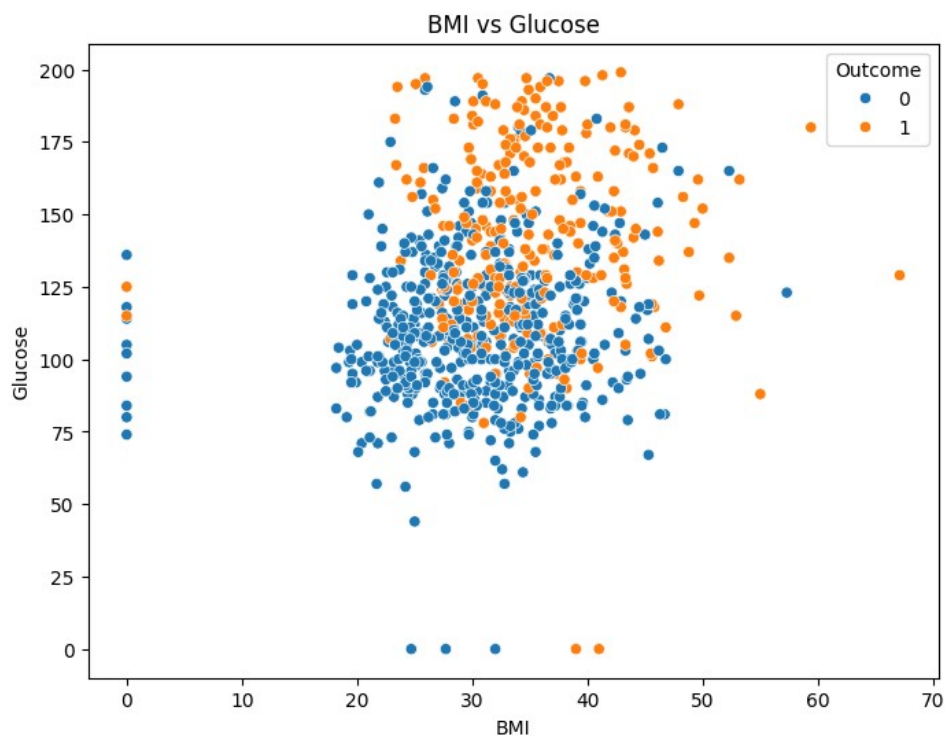


Figure 6: Scatter plot of BMI vs Glucose, colored by diabetes Outcome

Interpretation: Diabetic patients (orange) are visibly concentrated toward the upper-right region of the plot — i.e., higher Glucose and, to a lesser extent, higher BMI — while non-diabetic patients (blue) are more spread across lower-to-mid Glucose values. This visually supports Glucose as a strong discriminator between the two classes, with BMI contributing a secondary, weaker separation.

6.2 Age vs Glucose (colored by Outcome)



Figure 7: Scatter plot of Age vs Glucose, colored by diabetes Outcome

Interpretation: Diabetic cases appear across the full age range but are noticeably denser at higher Glucose levels regardless of age, while younger non-diabetic patients dominate the lower-Glucose region. This suggests Glucose remains a more consistent separator of Outcome than Age alone, although diabetes prevalence does appear to increase somewhat with age.

6.3 Pair Plot — All Features by Outcome



Figure 8: Pairwise relationships between all numeric features, colored by Outcome

Interpretation: Across the pair plot, Glucose shows the clearest separation between the two Outcome groups (visible in its diagonal KDE and in scatter panels involving Glucose), followed by BMI. Most other feature pairs show substantial overlap between diabetic and non-diabetic patients, indicating that no single variable other than Glucose cleanly separates the classes on its own — supporting the use of multiple combined features (as in a multivariate model) for more accurate prediction.

7. Correlation Analysis

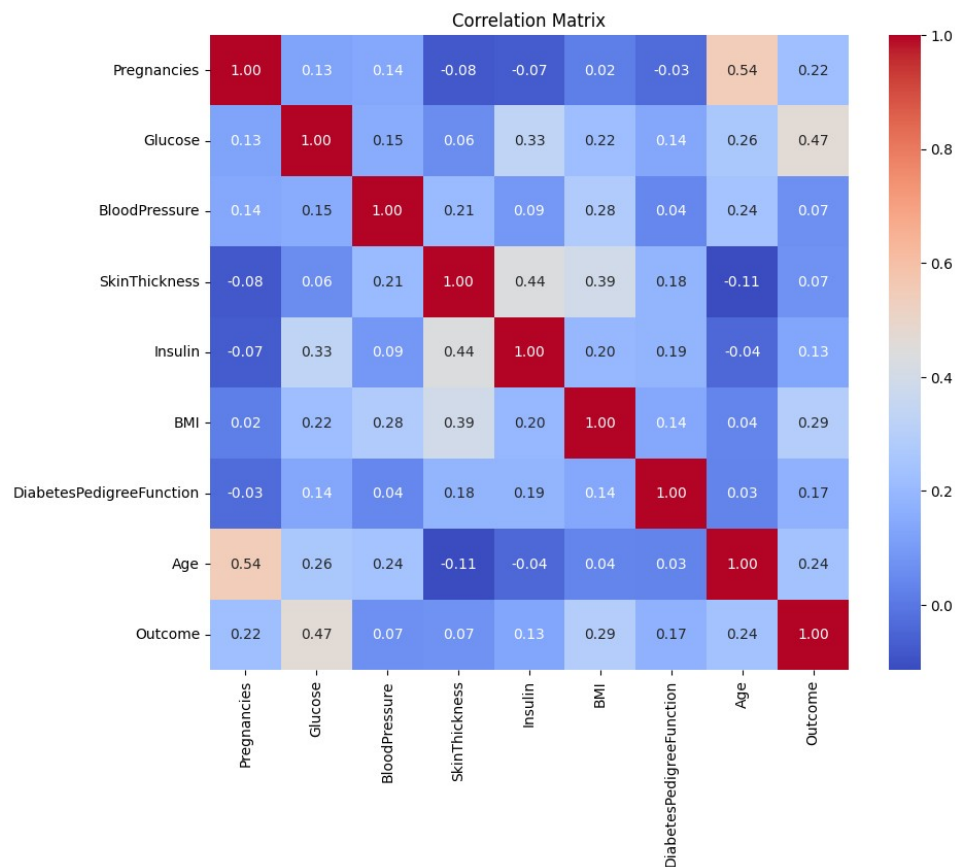


Figure 9: Correlation heatmap of all numeric features

Interpretation: Outcome correlates most strongly with Glucose ($r \approx 0.47$), followed by BMI ($r \approx 0.29$), Age ($r \approx 0.24$), and Pregnancies ($r \approx 0.22$) — all positive, moderate-to-weak correlations. BloodPressure and Insulin show weak correlations with Outcome ($r \approx 0.07$ and 0.13 respectively). Among predictors themselves, Age and Pregnancies are notably correlated ($r \approx 0.54$), and Insulin and SkinThickness are moderately correlated ($r \approx 0.44$), which is expected physiologically (older patients tend to have had more pregnancies; skin fold thickness relates to body fat and insulin resistance). No pair of predictors is so highly correlated as to suggest severe multicollinearity.

8. Class Balance of the Target Variable

8.1 Count Plot

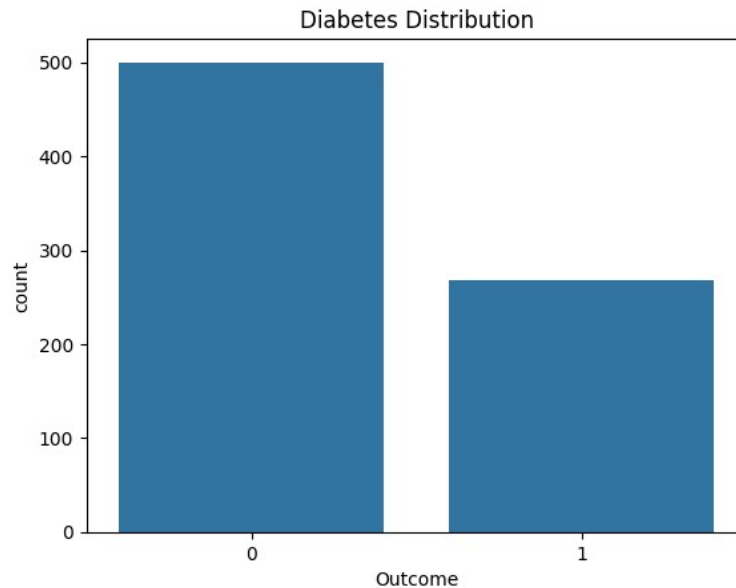


Figure 10: Count of patients by diabetes Outcome

8.2 Pie Chart

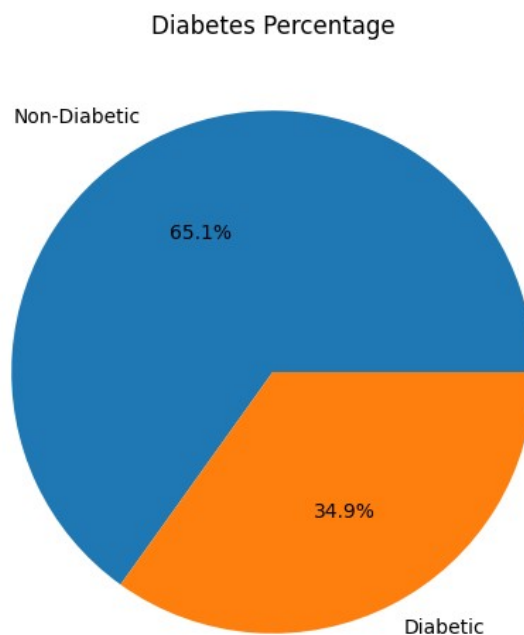


Figure 11: Percentage split of diabetic vs non-diabetic patients

Interpretation: The dataset is moderately imbalanced: 65.1% of patients are non-diabetic (Outcome = 0) versus 34.9% diabetic (Outcome = 1), a roughly 2:1 ratio. This imbalance is common in medical datasets and should be accounted for in any future classification model (e.g., via class weighting, resampling, or evaluation metrics such

as recall/F1-score rather than accuracy alone) to avoid a model that is biased toward predicting the majority (non-diabetic) class.

9. Glucose Distribution by Diabetes Status

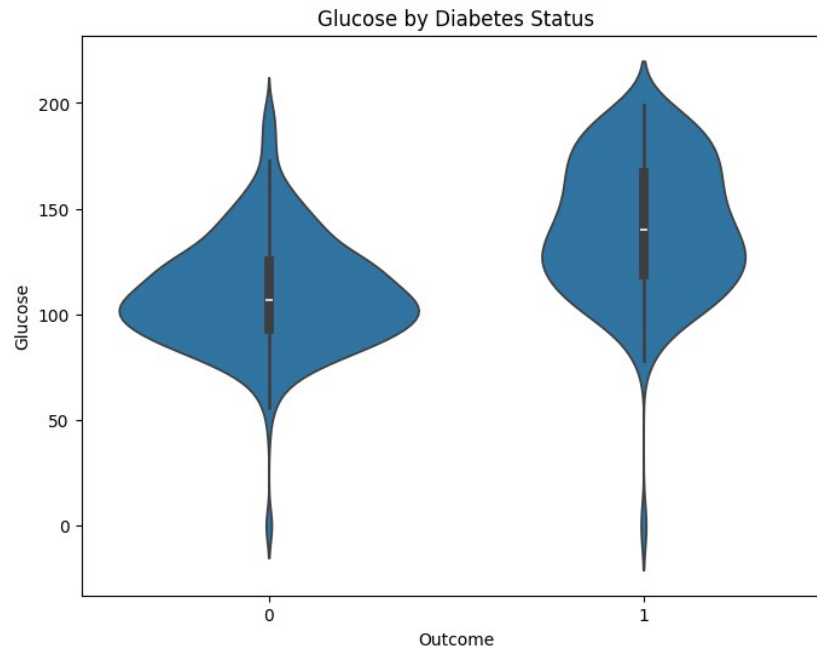


Figure 12: Violin plot of Glucose levels split by Outcome

Interpretation: The violin plot confirms what the correlation and scatter analyses suggested: diabetic patients (Outcome = 1) have a visibly higher median Glucose level (~140) and a distribution shifted right, compared to non-diabetic patients (Outcome = 0), whose median Glucose sits around ~105–110. The two distributions overlap in the middle range but are clearly offset, reinforcing Glucose as the single strongest visual and statistical marker of diabetes status in this dataset.

10. Statistical Hypothesis Testing

For each continuous variable below, an independent-samples t-test was used to test:

- Null Hypothesis (H_0): the mean of the variable is equal between diabetic and non-diabetic groups.
- Alternative Hypothesis (H_1): the mean of the variable differs between the two groups.

A significance level of $\alpha = 0.05$ was used throughout.

10.1 Group Means by Outcome

Variable	Non-Diabetic (0) Mean	Diabetic (1) Mean
Pregnancies	3.30	4.87
Glucose	109.98	141.26
BloodPressure	68.18	70.82
SkinThickness	19.66	22.16

Insulin	68.79	100.34
BMI	30.30	35.14
DiabetesPedigreeFunction	0.430	0.551
Age	31.19	37.07

10.2 Test Results

Variable	T-Statistic	P-Value	Decision ($\alpha = 0.05$)
Glucose	14.60	8.94×10^{-43}	Reject H_0 — significant
BMI	8.47	1.23×10^{-16}	Reject H_0 — significant
BloodPressure	1.80	0.0715	Fail to reject H_0 — not significant
Age	6.79	2.21×10^{-11}	Reject H_0 — significant
Pregnancies	6.30	5.07×10^{-10}	Reject H_0 — significant

Interpretation: Glucose shows by far the largest t-statistic and smallest p-value, confirming it as the most statistically significant predictor of diabetes status in this dataset — consistent with the visual evidence from the scatter, pair, and violin plots. BMI, Age, and Pregnancies are also statistically significant ($p < 0.001$), meaning their group-mean differences are very unlikely to be due to chance. BloodPressure, however, fails to reach significance at $\alpha = 0.05$ ($p = 0.072$), suggesting that diastolic blood pressure alone does not differ meaningfully between diabetic and non-diabetic patients in this sample — its earlier weak correlation with Outcome ($r \approx 0.07$) is therefore statistically corroborated.

10.3 One-Way ANOVA — Glucose Across Age Groups

Patients were split into three age bands (Under 30, 30–49, and 50+) to test whether mean Glucose differs across age groups.

Test	F-Statistic	P-Value	Decision ($\alpha = 0.05$)
One-Way ANOVA (Glucose ~ Age Group)	29.46	4.74×10^{-13}	Reject H_0 — significant

Interpretation: The ANOVA result is highly significant, indicating that mean Glucose levels differ significantly across at least one pair of the three age bands. Combined with the moderate positive correlation between Age and Outcome, this supports the general clinical understanding that glucose regulation tends to decline with age, making Age a useful supporting (though secondary) feature alongside Glucose.

10.4 Normality Check — Shapiro-Wilk Test on Glucose

A Shapiro-Wilk test was additionally run on the Glucose variable to formally assess normality (a common assumption check before applying parametric tests). The test was executed in the notebook using `stats.shapiro(df['Glucose'])`; given the large sample size ($n = 768$) common in this type of dataset, Shapiro-Wilk

tests typically detect even minor deviations from normality as statistically significant, so results should be interpreted alongside the histogram in Figure 1 rather than in isolation.

11. Key Findings & Conclusion

- Glucose is the single strongest predictor of diabetes in this dataset, both visually (histograms, scatter plots, violin plot) and statistically (highest t-statistic, smallest p-value, highest correlation with Outcome).
- BMI, Age, and Pregnancies are all statistically significant secondary predictors, each showing meaningfully higher averages among diabetic patients.
- BloodPressure does not show a statistically significant difference between diabetic and non-diabetic groups in this sample and is the weakest of the tested predictors.
- The dataset shows class imbalance (65.1% non-diabetic vs 34.9% diabetic), which should be addressed in any downstream predictive modeling.
- Several columns (Glucose, BloodPressure, SkinThickness, Insulin, BMI) contain biologically implausible zero values that likely represent missing data and should be cleaned/imputed before building a predictive model.
- Mean Glucose levels differ significantly across age bands, reinforcing Age as a relevant, if secondary, contextual factor.

Overall, this exploratory and statistical analysis identifies Glucose, BMI, Age, and Pregnancies as the most promising features for a future diabetes classification model, while flagging data-quality issues (disguised zeros) and class imbalance as important considerations for the modeling stage that would follow this EDA.

12. Suggested Next Steps

- Clean disguised missing values (zeros) in Glucose, BloodPressure, SkinThickness, Insulin, and BMI via imputation (e.g., median/group-median imputation).
- Address class imbalance using techniques such as class weighting, SMOTE, or stratified sampling.
- Build and evaluate classification models (e.g., Logistic Regression, Random Forest, XGBoost) using Glucose, BMI, Age, and Pregnancies as primary features.
- Evaluate models using recall, precision, F1-score, and ROC-AUC rather than accuracy alone, given the class imbalance.
- Publish the notebook, this report, and supporting files to a public GitHub repository (see README below) for reproducibility and portfolio presentation.

13. Project Repository (GitHub) — README Summary

A companion README.md file has been prepared for a GitHub repository housing this project's notebook, report, and visualizations. Its recommended structure is:

- Project title and one-line description
- Objective and dataset source/citation
- Repository structure (notebook, /images, /reports)
- Summary of methodology and key findings (as in Sections 2 and 11 above)
- Tools and libraries used

- How to run the notebook
- License and dataset attribution

Note: Because this assistant does not have the ability to create accounts or push to external services, the actual GitHub repository must be created and the files uploaded by you. The README.md file provided alongside this report is ready to be added directly to that repository — see the accompanying README.md file.